Private In-House Matrix & Element Deployment Guide

> **Executive Summary:** You can run a completely isolated, private Matrix server and customize the Element client exclusively for your company. Both Matrix and Element are open-source, allowing you to disable public federation and modify the software for internal use across Web, Desktop, Mobile, and Terminal platforms with Active Directory/ LDAP Single Sign-On (SSO).

Part 1: Server Isolation & Client Customization

Locking Down the Server (No Public Users)

You do not need to rewrite the Matrix protocol to keep outsiders out. Standard Matrix homeservers (such as Synapse or Dendrite) include built-in configuration flags to isolate your deployment:

- **Disable Federation.** Turn off federation in your server configuration (for example, setting `federation_domain_whitelist: []` in Synapse) so your server refuses to communicate with `matrix.org` or any external Matrix servers.
- **Restrict Registration.** Disable public account registration and connect the server to your company's internal Single Sign-On (SSO), LDAP, or Active Directory so only verified employees can log in.
- **Air-Gapped Hosting.** Host the entire backend on your internal corporate network or VPN so the server is never reachable from the public internet.
- **Distributed Internal Servers.** If your company has multiple offices or departments running separate Matrix servers, whitelist *only* your internal domains so they federate with each other while blocking external servers.

Modifying the Element Client

Because the Element Open Source codebase is publicly available, you have full freedom to customize it for your team:

- **Custom Branding and UI.** Fork Element Web, Desktop, or Mobile to apply your company's logos, colors, and custom domain defaults.
- **Locking the Homeserver URL.** Edit the client's `config.json` file to hardcode your internal server URL and hide the option to switch to public servers.
- **Feature Stripping.** Disable public room directories, external identity servers, and third-party integrations directly through configuration files or code modifications.
- **Licensing Note.** Element clients are generally licensed under Apache 2.0 or AGPLv3 (depending on the specific repository). If you modify an AGPLv3 component and distribute it or host it as a network service, verify the license terms to ensure compliance with source-sharing requirements for your users.

Part 2: Multi-Platform & Active Directory / LDAP Architecture

You can cover Web, Desktop, iOS, Android, and Terminal across Linux, BSD, GNU/Hurd, macOS, and Windows while integrating your internal Active Directory/LDAP server for SSO. Because Matrix separates the server protocol from client applications, you do not have to build or maintain every client from scratch.

1. Integrating Active Directory / LDAP for Company SSO

Modern Matrix deployments handle authentication using the **Matrix Authentication Service (MAS)**, which relies on OpenID Connect (OIDC). To connect your internal directory across both GUI and terminal clients, use the following architecture:

- **Bridge LDAP to OIDC.** Because MAS speaks OIDC rather than raw LDAP, place a lightweight open-source OIDC identity provider like **Dex** or **Keycloak** in front of your Active Directory or LDAP server. This translates LDAP credentials into standard OIDC SSO tokens for Matrix.
- **Support Terminal Clients via Device Authorization.** Terminal clients do not have an embedded web browser to render an SSO login screen. By enabling the **OAuth 2.0 Device Authorization Grant** in MAS, terminal users receive a short URL and code to approve their login in their desktop browser using their regular LDAP SSO session.
- **Legacy Direct LDAP Option.** If you use an older Synapse setup without MAS, you can use the `matrix-synapse-ldap3` module to authenticate directly against LDAP, though MAS with Dex is the current standard for newer clients like Element X.

2. Deploying Clients Across All Platforms

Combine customized Element builds for standard devices with native open-source clients for Unix-like terminals:

Platform Category	Recommended Client	Customization & Deployment Approach
Web & Desktop (Linux, macOS, Windows)	**Element Web / Element Desktop**	Fork and build with a custom `config.json` that hardcodes your homeserver URL, sets your company branding, and disables public server discovery.
Mobile (iOS & Android)	**Element X**	Rebrand the open-source mobile repositories and distribute internal builds via your company's Mobile Device Management (MDM) or enterprise app store.
Terminal (Linux, BSD, macOS, Windows)	**iamb** or **gomuks**	Both are open-source terminal clients supporting end-to-end encryption (E2EE) and are available in standard Linux and BSD package repositories.
GNU/Hurd	**gomuks** or **WeeChat-Matrix**	Because Hurd lacks official Rust/Node targets for Element, compile **gomuks** (written in Go) from source or run the Python-based `weechat-matrix` plugin inside WeeChat.

3. Enforcing Total Isolation Across Every Client

To guarantee that no employee accidentally leaks messages or connects to outside servers regardless of which client they use, enforce isolation at the server and network layers rather than relying solely on client UI tweaks:

- **Server-Side Federation Block.** Set `federation_domain_whitelist: []` in your Synapse `homeserver.yaml` file. Even if a user runs an unmodified terminal client on Linux or BSD, the server will refuse to send or receive messages from any external Matrix domain.
- **Network-Level Firewalling.** Place your Synapse server, MAS, and LDAP/OIDC bridge behind your corporate firewall or WireGuard/OpenVPN mesh so the endpoints are completely unreachable from the public internet.
- **Pre-Configured Client Profiles.** Distribute a pre-filled configuration file (such as `config.toml` for `iamb` or `config.json` for Element) via your internal configuration management tools (like Ansible or Puppet) so employees only have to run the app and sign in with their LDAP credentials.